



MXWneXt

Command Strings

MXW neXt Command Strings
Version: 1.0 (2024-C)

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MXWneXt Command Strings

TPCI Overview

MXW neXt devices are connected via Ethernet to a control system, such as

- AMX, Crestron or Extron
- Symetrix, Biamp, other digital signal processors (DSP)
- Specialized custom programs

Connection: Ethernet (TCP/IP; select “Client” in the AMX/Crestron program)

Port: 2202

MXW neXt Naming

- MXW1X - hybrid bodypack transmitter
- MXW2X - handheld transmitter
- MXW6X - boundary transmitter
- MXWAPXD2 - 2-channel access point dock

Conventions

There are 3 types of strings:

GET	Finds the status of a property. After the AMX/Crestron sends a GET command, the system responds with a report (REP) string
SET	Changes the status of a property. After the AMX/Crestron sends a SET command, the system responds with a report (REP) string to indicate the new value of the property.
REP	When the system receives a GET or SET command, it replies with a report (REP) string to indicate the status of the property.

All messages sent and received using a restricted set of printable characters (ASCII-7).

Most parameters will send an acknowledgement in the form of a **REP** string when they change. Thus, it is not necessary to constantly query status parameters. The MXWAPXD2 will send a **REP** string when any of these parameters change. All commands are sent back and forth to the MXWAPXD2. Transmitter commands are sent to the MXWAPXD2, which then forwards them to the transmitter. Thus, for control, simply send commands to the control IP address associated with the MXWAPXD2.

MXWAPXD2 Commands

Get All

Parameter Name:	ALL
Command Types Supported:	GET, REP
Indexing:	n/a
Value(s):	Responds with REP for all device-specific properties and ALL channel-related properties.
Example(s):	< GET ALL >

Automixer Out Mute State

Parameter Name:	AUTOMIX_OUT_MUTE
Command Types Supported:	GET, SET, REP
Indexing:	n/a
Value(s):	status refers to the audio state: MUTED means the automixer output is muted ACTIVE means the automixer output is unmuted
Example(s):	< GET AUTOMIX_OUT_MUTE > < REP AUTOMIX_OUT_MUTE status > < SET AUTOMIX_OUT_MUTE status > < REP AUTOMIX_OUT_MUTE status >

System Preset Mode

Parameter Name:	SYSTEM_PRESET_MODE
Command Types Supported:	GET, SET, REP
Indexing:	n/a
Value(s):	Where mode is one of the allowed MXW neXt preset modes: PRESENTATION CONFERENCE DIRECT

	ERR: Invalid preset mode indicated
Example(s):	<pre>< GET SYSTEM_PRESET_MODE > < REP SYSTEM_PRESET_MODE mode > < SET SYSTEM_PRESET_MODE mode > < REP SYSTEM_PRESET_MODE mode ></pre>

Input Channel Gain

Parameter Name:	INPUT_CH_GAIN																														
Command Types Supported:	GET, SET, REP																														
Indexing:	channel number, see Value(s)																														
Value(s):	Where x is the channel number: 1: Mic 1 In 2: Mic 2 In 3: Analog In 4: Dante 1 In 5: Dante 2 In 6: USB In Where y is the input gain value, with a range of 0 – 1400, representing gain from -110.0 dB to +30.0 dB in steps of 0.1 dB. Gain multiplied by 10 and offset by +1100 (y = [gain] × 10 + 1100). Default gain value is set to 1100 (0 dB). <table> <tr><td>+30 dB –</td><td>– 1400</td></tr> <tr><td>+20 dB –</td><td>– 1300</td></tr> <tr><td>+10 dB –</td><td>– 1200</td></tr> <tr><td>0 dB –</td><td>– 1100</td></tr> <tr><td>-10 dB –</td><td>– 1000</td></tr> <tr><td>-20 dB –</td><td>– 900</td></tr> <tr><td>-30 dB –</td><td>– 800</td></tr> <tr><td>-40 dB –</td><td>– 700</td></tr> <tr><td>-50 dB –</td><td>– 600</td></tr> <tr><td>-60 dB –</td><td>– 500</td></tr> <tr><td>-70 dB –</td><td>– 400</td></tr> <tr><td>-80 dB –</td><td>– 300</td></tr> <tr><td>-90 dB –</td><td>– 200</td></tr> <tr><td>-100 dB –</td><td>– 100</td></tr> <tr><td>-110 dB –</td><td>– 0</td></tr> </table>	+30 dB –	– 1400	+20 dB –	– 1300	+10 dB –	– 1200	0 dB –	– 1100	-10 dB –	– 1000	-20 dB –	– 900	-30 dB –	– 800	-40 dB –	– 700	-50 dB –	– 600	-60 dB –	– 500	-70 dB –	– 400	-80 dB –	– 300	-90 dB –	– 200	-100 dB –	– 100	-110 dB –	– 0
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-70 dB –	– 400																														
-80 dB –	– 300																														
-90 dB –	– 200																														
-100 dB –	– 100																														
-110 dB –	– 0																														
Example(s):	<pre>< GET x INPUT_CH_GAIN > < SET x INPUT_CH_GAIN y > < REP x INPUT_CH_GAIN y ></pre>																														

Retrieve Firmware Version

Parameter Name:	FW_VER
Command Types Supported:	GET, REP
Indexing:	n/a
Value(s):	Format: 24 character string. Results are padded with spaces to ensure 24 characters are always reported. Package version number presented in dot-decimal notation: Maj.Min.Pack.Build * = {SelfTestFailed}
Example(s):	Example where self test passed: <pre>< GET FW_VER > < REP FW_VER {2.0.15.2 } ></pre> Example where self test failed: <pre>< GET FW_VER > < REP FW_VER {2.0.15.2* } ></pre>

Serial Number

Parameter Name:	SERIAL_NUM
Command Types Supported:	GET, REP
Indexing:	n/a
Value(s):	serial_num is an 11-character alphanumeric string.
Example(s):	<pre>< GET SERIAL_NUM > < REP SERIAL_NUM {serial_num } ></pre>

Reboot

Note: This command does not send acknowledgement.

Parameter Name:	REBOOT
Command Types Supported:	SET
Indexing:	n/a

Value(s):	n/a
Example(s):	< SET REBOOT >

Device ID

Parameter Name:	DEVICE_ID
Command Types Supported:	GET, SET, REP
Indexing:	n/a
Value(s):	device_id is a string of 1 to 31 alphanumeric characters from the set A-Z, a-z, 0-9, and '-' (dash or hyphen). MXWAPXD2 names must be unique on the network, and follow domain name service (DNS) naming rules. Names are not case-sensitive; “Guitar” and “guitar” are treated as the same label. Unicode and non-roman characters are not supported. Device names cannot begin or end with a hyphen or dash. Names longer than 31 characters are truncated to 31. There is no need to pad the name out to 31 characters with spaces when setting a name.
Example(s):	<pre>< GET DEVICE_ID > < REP DEVICE_ID {device_id} > < SET DEVICE_ID {device_id} > < REP DEVICE_ID {device_id} ></pre>

Transmitter Commands

These commands are to be sent to the MXWAPX IP address.

Transmitter Status

Parameter Name:	TX_STATUS
Command Types Supported:	GET, REP
Indexing:	bay or channel number, see Value(s)
Value(s):	bay/ch defines whether this command is aimed at a docking bay (BAY) or a channel (CH) x is the bay or channel number requested, 0 for GET all

	sts is the transmitter status: ACTIVE: Transmitter is undocked, powered on, unmuted MUTED: Transmitter is undocked, powered on muted OFF: Transmitter status set to off (SET only, no report is received) ON_CHARGER: Transmitter is docked in the bay (REP only) UNKNOWN: Transmitter is offline/unlinked (REP only) If transmitter is SET to OFF, any further SET commands will yield UNKNOWN report.
Example(s):	<pre>< GET bay/ch x TX_STATUS > < SET bay/ch x TX_STATUS sts > < REP bay/ch x TX_STATUS sts ></pre>

Transmitter Device ID

Parameter Name:	TX_DEVICE_ID
Command Types Supported:	GET, SET, REP
Indexing:	bay or channel number, see Value(s)
Value(s):	device_id is a string of 1 to 12 alphanumeric characters from the set A-Z, a-z, 0-9, and '-' (dash or hyphen). Transmitter names cannot begin or end with a hyphen or dash. Names are padded with spaces to ensure 12 characters are always reported. There is no need to pad the name out to 12 characters with spaces when setting a name. Names longer than 12 characters result in ERR. bay/ch defines whether this command is aimed at a docking bay (BAY) or a channel (CH). x is the bay or channel number requested, 0 for GET all. UNKNOWN: No transmitter is linked or transmitter is off. ERR: x does not match a valid channel or docking bay
Example(s):	<pre>< GET bay/ch x TX_DEVICE_ID > < REP bay/ch x TX_DEVICE_ID {device_id} > < SET bay/ch x TX_DEVICE_ID {device_id} > < REP bay/ch x TX_DEVICE_ID {device_id} ></pre>

Transmitter Model

Parameter Name:	TX_MODEL
Command Types Supported:	GET, REP
Indexing:	bay or channel number, see Value(s)
Value(s):	bay/ch defines whether this command is aimed at a docking bay (BAY) or a channel (CH)

	x is the bay or channel number requested, 0 for GET all. model is one of the allowed MXW neXt TX_MODEL values: MXW1X MXW2X MXW6X UNKNOWN: No transmitter is linked or transmitter is off. ERR: x does not match a valid channel or docking bay
Example(s):	< GET bay/ch x TX_MODEL > < REP bay/ch x TX_MODEL model > < REP bay/ch x TX_MODEL UNKNOWN >